

Titanic Dataset Data Cleaning & Visualization Project

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Internship: Data Science Internship

Task: Task 1 – Data Cleaning & Visualization Project

1.Introduction

Data cleaning and visualization are important steps in data science. This project uses the Titanic dataset to demonstrate data preprocessing, handling missing values, and generating visual insights using Python.

1.1 Objective

The objective of this project is to clean a raw dataset, handle missing values and duplicates, and perform data visualization to gain meaningful insights. The Titanic dataset was used for this analysis.

2. Dataset Description

The Titanic dataset contains information about passengers who traveled on the Titanic ship. The dataset includes various features such as passenger class, age, gender, fare, survival status, and other details.

Dataset Size: 891 rows and 12 columns.

Features Used

- Passenger Id
- Survived
- P class
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

3. Tools and Technologies Used

- Python

- Google Colab
- Pandas
- Matplotlib
- Seaborn

4. Data Cleaning Process

The following data cleaning operations were performed:

4.1 Handling Missing Values

- Missing values in the Age column were replaced using the median value.
- Missing values in the Embarked column were replaced using the mode value.

4.2 Removing Duplicates

- Duplicate records were identified and removed from the dataset.

4.3 Data Verification

- The dataset was checked again after cleaning to ensure there were no significant missing values affecting the analysis.

5. Data Visualization

The following visualizations were created:

5.1 Survival Count

A count plot was used to visualize the number of passengers who survived and those who did not survive.

5.2 Gender vs Survival

A comparison chart was created to analyze survival rates among male and female passengers.

5.3 Age Distribution

A histogram was used to study the age distribution of passengers.

5.4 Passenger Class Distribution

A count plot was created to analyze the number of passengers in each travel class.

6. Key Findings

1. The number of passengers who did not survive was greater than the number of survivors.
2. Female passengers had a significantly higher survival rate than male passengers.
3. Most passengers were between 20 and 40 years of age.
4. Third-class passengers represented the largest group of travelers.
5. Passenger class had a noticeable impact on survival rates.

7. Conclusion

The Titanic dataset was successfully cleaned and analyzed using Python. Missing values were handled, duplicate records were removed, and several visualizations were created to understand the data better. The project demonstrates fundamental data preprocessing and visualization techniques commonly used in data analytics.

This project helped in understanding the fundamentals of Exploratory Data Analysis (EDA) and data visualization techniques used in Data Science.

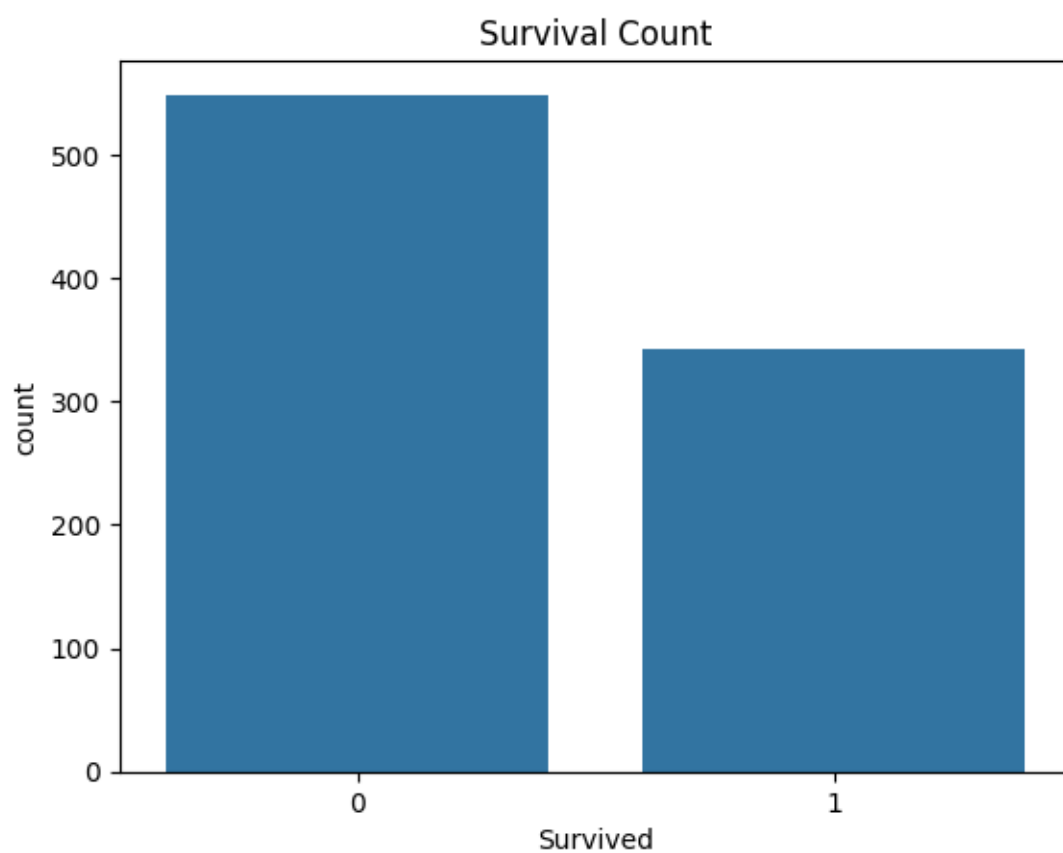
8. References

- Titanic Dataset
- Python Documentation
- Pandas Documentation

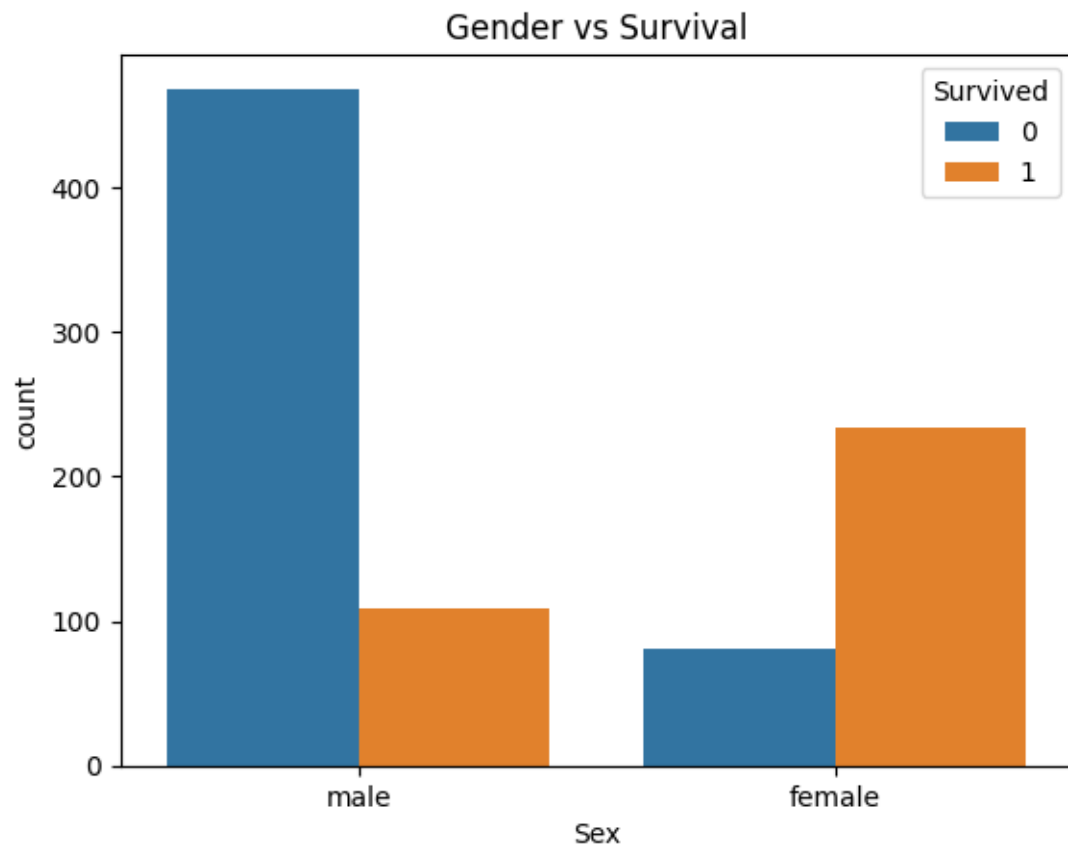
- Matplotlib Documentation
- Seaborn Documentation

9. Output Screenshots

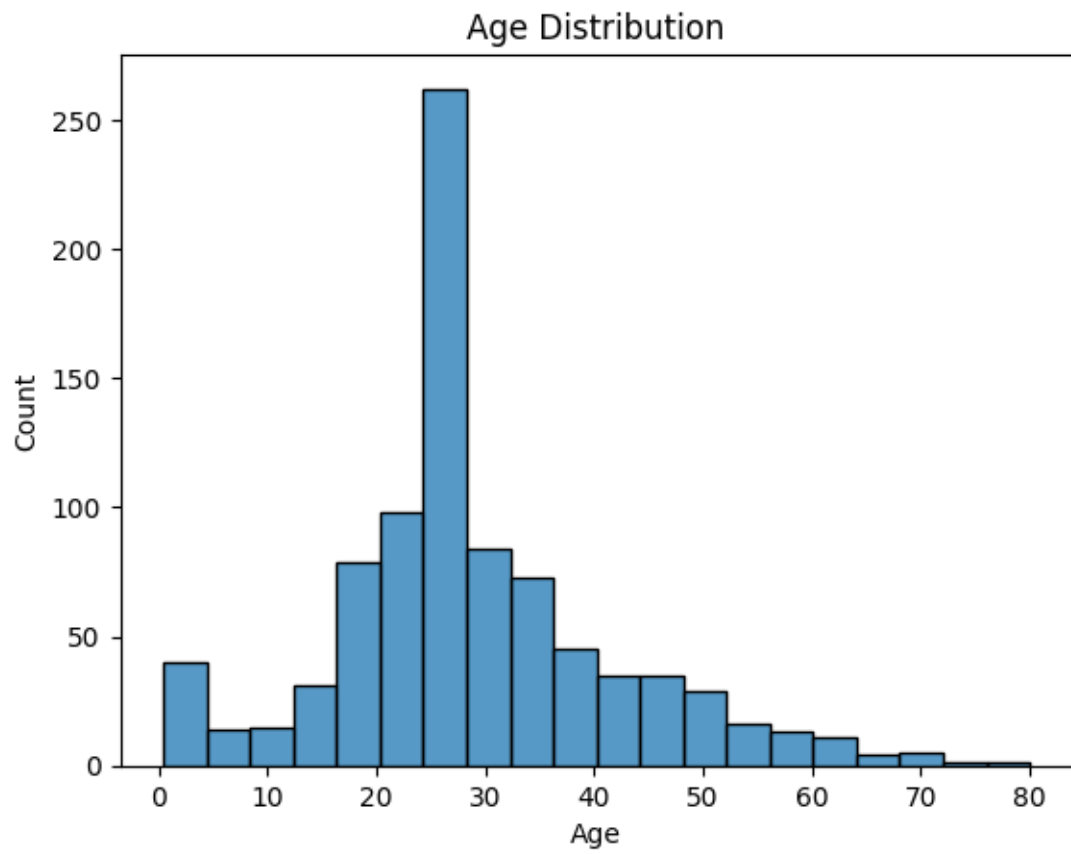
1.Survival Count Graph



2. Gender vs Survival Graph



3. Age Distribution Graph



4. Passenger Class Distribution Graph

